

Designing for Sustained Engagement in Anxiety Disorders: Lessons from MEMOPAD's Co-Design Journey

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1 INTRODUCTION

Digital mental health interventions (DMHIs) face a well-documented paradox: while efficacy is demonstrated in controlled trials, real-world effectiveness remains constrained by high attrition rates, a problem characterized as the "Achilles heel" of digital therapeutics [1]. A recent meta-analysis of 176 randomized controlled trials found that mental health apps show promise for depression and anxiety symptoms, yet engagement typically collapses within weeks of initial use [2]. Anxiety disorders, including generalized anxiety disorder (GAD), panic disorder, social anxiety disorder, obsessive-compulsive disorder (OCD), and post-traumatic stress disorder (PTSD), present population-specific engagement barriers. The symptoms these tools aim to address (including avoidance behaviours, cognitive overload, fear of judgment) may actively work against sustained use. Standard self-reported approaches rely paradoxically on patients' ability to assess their own emotional awareness, a capability often impaired in anxiety disorders [3].

Co-design has been proposed as one approach to address the dropout problem by involving end-users in development [4, 5]. However, few studies document how co-design with anxiety populations specifically shaped engagement-focused design decisions. This use case presents lessons from MEMOPAD (Multimodal Emotion Monitoring in Clinical Pathways for Anxiety Disorders), a 6+1 phase co-design process (currently 5.5 completed) ranging from analyzing the problem and

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needs of the different stakeholders till developing a wearable emotion monitoring system. With 19-25 participants per phase, including people with lived experience, unpaid carers, and clinicians from NHS and private sector settings, three intervention scenarios emerged: (a) improving clinical conversations, (b) breaking anxiety loops, and (c) supporting timely help-seeking behaviour.

2 HOW WE DEFINED ENGAGEMENT

A key finding was participants' consistent request to focus on the full emotion spectrum, not exclusively anxiety or negative states. This anti-stigma design principle emerged as potentially central to sustained engagement: participants wanted a tool reflecting their complete emotional lives rather than one that pathologized their experiences. Phase 0 consultations revealed that patients advocated for including carers in the co-design process itself. This suggests that engagement design for anxiety disorders may need to address the broader care ecosystem rather than treating users as isolated individuals. The three dimensions of engagement considered in MEMOPAD are reported in Table 1.

Table 1. Three dimensions of engagement emerged inductively from co-design activities

Dimension	Operationalization	Examples of Design Implication
Emotional	Interaction perceived as non-threatening and stigmatizing	Frame experiences through emotion zones rather than disorder-specific language
Cognitive	System reduces rather than adds mental load	Context scaffolding, simplified classification, guided semi-structured self-reporting
Social	Connection to care ecosystem without perceived surveillance	Carer involvement, optional data sharing, peer support feature

3 HOW WE VALIDATED ENGAGEMENT

Co-design methodology: Six phases plus initial consultations (Phase 0), with 19-25 participants per phase including people with lived experience of anxiety disorders, unpaid carers, and clinicians. Each phase commenced with presentation of prior phase findings, enabling iterative refinement.

Cross-cohort validation: Phase V involved 16 participants in prototype testing under lab conditions. Notably, 14 participants were new to MEMOPAD, yet they confirmed appreciation for features prioritized by earlier co-design cohorts. This suggests the engagement strategies may generalize beyond the original co-design participants. A larger sample of participants will be invited to express their feedback via online surveys.

Methodological observation: Synchronous co-design tools (e.g. collaborative digital whiteboards) proved counter-productive for discussions with the public, risking triggering anxiety during the co-design sessions. Lower-pressure, asynchronous alternatives increased participation and appeared more inclusive.

4 HOW WE DESIGNED FOR ENGAGEMENT

The system is conceptualised to combine continuous physiological monitoring (electrodermal activity, heart rate, skin temperature via commercially available wearable devices like Google Pixel Watch 2) with user-generated self-reports. This parallel data stream approach emerged from co-design discussions balancing the value of passive detection against concerns about autonomy. Across the five co-design phases completed to date, several potential engagement barriers have been identified, along with approaches to mitigate them, as reported in Table 2.

Table 2. Identified engagement barriers

Engagement Barrier	Design Strategy	Implemented Feature
Anxiety-related avoidance	Low-threat entry points	4-color emotion zones replacing clinical terminology; character-based guidance
Initiation difficulty	Context scaffolding	System pre-populates context from passive data; flexible semi-structured self-report forms
Notification fatigue	Adaptive prompting	User-controlled notification frequency, timing, and content preferences
Gamification concerns	Non-competitive rewards	Points collection and badges without leaderboards or social comparison
Social isolation	Ecosystem connection	In-app private messaging; optional data sharing with carers and clinicians
Information overload	Multi-modal presentation	Hybrid visualization combining charts with RAG-generated narratives
Acute anxiety episodes	In-moment intervention	Smartwatch-based "gentle nudges" to interrupt anxiety loops
Deficit-focused framing	Full spectrum approach	Equal tracking of positive and negative emotional states

5 LESSONS LEARNED

- **Carers cannot be ignored:** Engagement may be relational rather than individual. Patients advocated for carer involvement from Phase 0 consultations. This suggests that designing for isolated users may miss how anxiety disorders are managed in daily life. MEMOPAD recognizes carers as users of the tool ecosystem.
- **Clinicians lack time:** Role-appropriate engagement pathways are necessary. Clinician dropout during co-design activities highlighted that engagement strategies must account for resource constraints. We adapted by implementing "low-touch" pathways for time-poor experts: 20-30 min online one-to-one interviews, an approach reflected in the user experience we developed for them in MEMOPAD ecosystem.
- **Emotions over anxiety:** Anti-stigma framing may support sustained engagement. Participants consistently requested tools reflecting their complete emotional lives rather than monitoring pathology. The 4-color emotion zones and full-spectrum tracking emerged as potentially important engagement enablers.
- **Features tailored for anxiety:** Population-specific barriers require population-aware solutions. Context scaffolding, gentle nudges, guided self-reporting, and extensive personalization options all address ways that anxiety symptoms may specifically undermine engagement with digital tools.

6 OPEN QUESTION FOR WORKSHOP DISCUSSION

DHIs can face dual failure modes: dropout (users abandon the tool before deriving benefit) and dependency (users become over-reliant). MEMOPAD's non-competitive gamification and carer-integration features attempt to navigate this tension, but frameworks for distinguishing sustained healthy engagement from problematic attachment remain underdeveloped. How might we operationalise and measure the difference between "engaged & empowered" VS "engaged & dependent"?

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